

Database Management Systems — Course Syllabus

Unit 1: Introduction to Databases

A database is an organized collection of structured data stored electronically. A Database Management System (DBMS) is software that manages the creation, retrieval, update, and deletion of data in a database while ensuring data integrity and security. Key advantages of using a DBMS over traditional file systems include reduced data redundancy, improved data consistency, easier data access through query languages, and better data security through access control mechanisms.

Unit 2: Relational Model and Normalization

The relational model organizes data into tables (relations) consisting of rows (tuples) and columns (attributes). Normalization is the process of organizing data in a database to reduce redundancy and improve data integrity. It involves decomposing tables into smaller, well-structured tables without losing information. The main normal forms are First Normal Form (1NF), which requires atomic column values; Second Normal Form (2NF), which removes partial dependencies on a composite key; Third Normal Form (3NF), which removes transitive dependencies; and Boyce-Codd Normal Form (BCNF), a stricter version of 3NF.

Unit 3: Transactions and Concurrency Control

A transaction is a logical unit of database processing that must maintain the ACID properties: Atomicity ensures a transaction is all-or-nothing; Consistency ensures the database moves from one valid state to another; Isolation ensures concurrent transactions do not interfere with each other; and Durability ensures committed changes survive system failures. Concurrency control techniques such as locking protocols and timestamp ordering are used to manage simultaneous access to data by multiple transactions while preserving these properties.

Unit 4: Indexing and Query Optimization

Indexes are data structures, commonly B-trees or hash tables, that improve the speed of data retrieval operations on a database table. Query optimization is the process by which a DBMS chooses the most efficient way to execute a given SQL query by evaluating alternative execution plans and estimating their costs based on factors such as available indexes, table sizes, and join order.